

Saurabh Tiwari

Software & Data Engineering

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience developing full-stack applications, REST APIs, data pipelines, and analytics dashboards using Python, Java, React, Node.js, SQL, and PostgreSQL. Built projects involving backend development, authentication, data validation, ETL workflows, data modeling, and real-time application features. Strong foundation in Data Structures & Algorithms, OOP, DBMS, and software engineering, with a focus on building scalable, maintainable technology solutions and applying analytical problem-solving to real-world business needs.

EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar
Bachelor of Technology (B.Tech) in Computer Science Engineering

Bhubaneswar, India
Graduation: 2027

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript

Full-Stack Development: React.js, Node.js, Express.js, FastAPI, REST APIs, JWT, Socket.io

Data Engineering & Analytics: ETL Pipelines, Pandas, NumPy, SQL, Data Validation, Schema Enforcement, Batch Processing, Streamlit

Databases: PostgreSQL, MySQL, MongoDB, SQLite

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, SDLC

Web & Tools: Vite, Tailwind CSS, HTML5, CSS3, Cloudinary, GitHub

PROJECTS

CampusKart – Full-Stack Campus Marketplace Platform

July 2026

React (Vite), Node.js, Express.js, MongoDB, Socket.io, JWT, Cloudinary, Tailwind CSS

- Architected and developed a full-stack marketplace with 9 RESTful API modules and 11 end-to-end features covering listings, messaging, wishlist, reviews, and rental workflows.
- Implemented JWT-based authentication, campus-domain access restriction, server-side authorization, and request validation across protected routes, resolving 10+ authorization and validation issues before launch.
- Engineered server-side search and filtering across 4 dimensions and real-time messaging with Socket.io supporting 50+ concurrent sessions; added rule-based validation to flag potentially underpriced listings.

GitHub: github.com/saurabhtiwari021/CampusKart **Live:** campus-kart-one.vercel.app

Internal Marking System – Academic Data Engineering & BI Platform

March 2026

Python, SQL, FastAPI, Streamlit, Pandas, PostgreSQL/SQLite, REST APIs

- Built an end-to-end academic data platform using Python, SQL, FastAPI, PostgreSQL, and Streamlit for data ingestion, transformation, validation, analysis, and reporting.
- Engineered a Pandas-based ETL and validation pipeline processing batches of 500+ student records across 5+ validation rules covering schemas, scores, IDs, dates, and duplicates; quarantined invalid records with detailed error tracking, reducing estimated manual data-cleanup effort by 70%.
- Designed SQL-driven analytics dashboards with subject-wise analysis, grade-distribution visualizations, and CSV exports, while enforcing API-key authentication and upload/CORS controls across data-ingestion endpoints.

GitHub: github.com/saurabhtiwari021/internal-marking-system **Live:** app-marking-system.streamlit.app

CERTIFICATIONS

Data Engineering - ExcelR, in association with KIIT University

Feb – Mar 2026

90-hour professional training program; Certificate ID: 55872/ExcelR [Certificate](#)

EXTRACURRICULAR ACTIVITIES

- Social Activist – TPH Society, 2025 – present
- Student Volunteer – Bakul Foundation, 2025